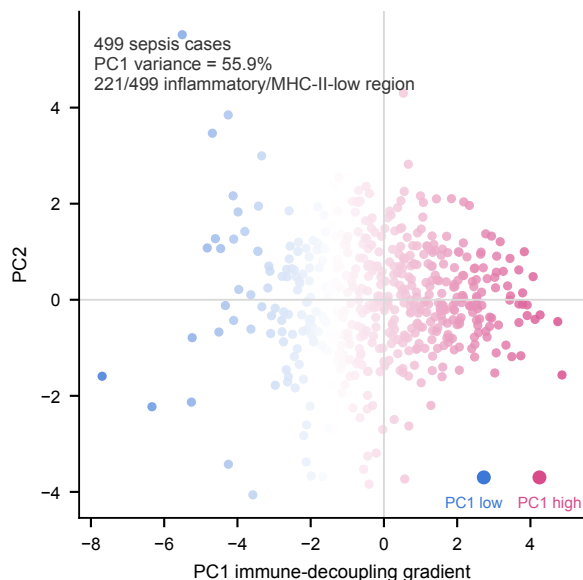
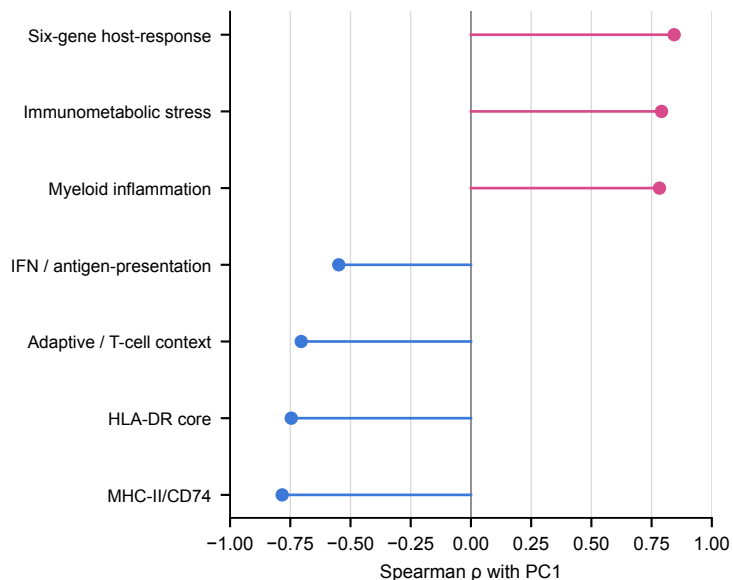


Figure 3. Continuous-spectrum immune decoupling and pathway programs.

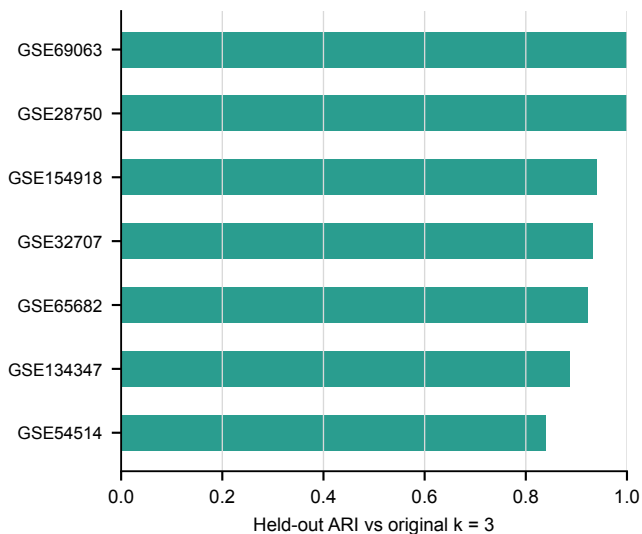
A Case-only immune-decoupling spectrum



B Signature alignment with PC1



C Cross-cohort transfer stability



D Signed pathway association

